

WEB TECHNOLOGIES

CSE-4004



LECTURE 1: DHTML, JavaScript



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DHTML (Dynamic HTML)

- **DHTML= HTML + CSS + JavaScript + DOM**
- **Used to create interactive and dynamic web pages.**
- **Change the content dynamically without loading it.**



HTML → **Structure of the Webpage**

CSS → **Styling and layout**



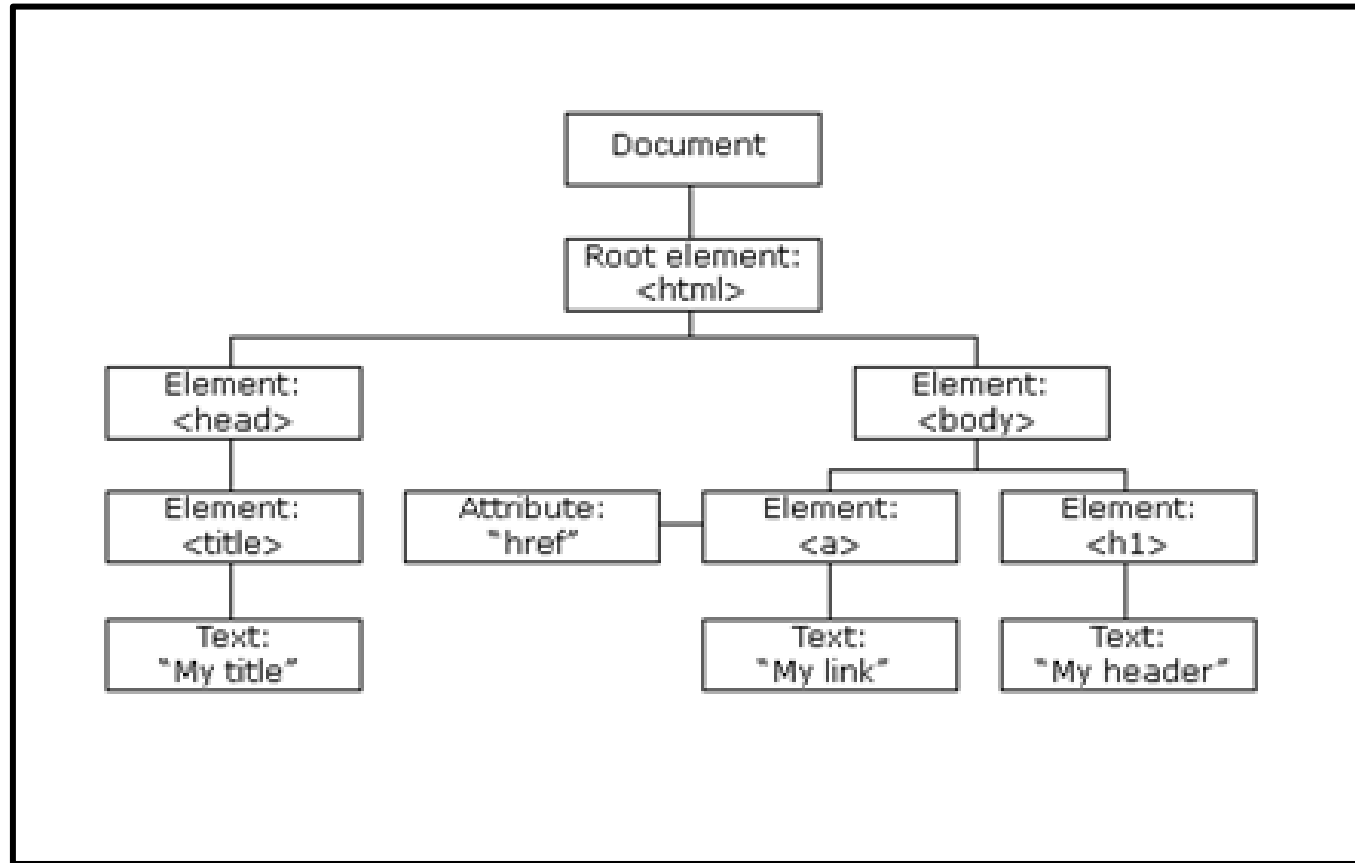
JavaScript → **Add interactivity and behavior to a web page.**

Introduction to JavaScript

- *JavaScript was invented by **Brendan Eich** in **1995** at **Netscape Communications**.*
- *Old name was **Mocha** later renamed to **LiveScript** and then renamed to **JavaScript**.*

DOCUMENT OBJECT MODEL (DOM)

DOM is a tree-like representation of an HTML document that allows JavaScript to interact with web page elements.



Features of JavaScript

- Interpreted
 - Client-side Scripting
 - Object Based
 - Event-Driven
 - Dynamic Typing
 - Platform Independent
 - Case Sensitive
 - Supports Functional Programming
 - Secure
- Used to create dynamic and interactive web pages.

What We Can Do With JavaScript

- Create Interactive Web Pages
- Manipulate HTML and CSS - Live Webpage
- Form Validation
- Handle Events
- Communicate with Servers
- Store data in browser
- Build Full Web Applications
- Create Games
- Create Mobile and Desktop Apps

➤ JS code is written inside **<SCRIPT> </SCRIPT>** tags.

➤ JS can "display" data in different ways:

i. Writing into an HTML element,
using **innerHTML** or **innerText**.

document.getElementById("demo").innerHTML = "Hello";

ii. Writing into the HTML output using **document.write()**.
document.write("Hello")

iii. Writing into an alert box, using **window.alert()**.

iv. Writing into the browser console, using **console.log()**.

Identifiers : Sequence of characters used to identify a variable, function or an object.

- The rules for constructing names (identifiers) are:
- Names can contain letters, digits, underscores, and dollar signs.
- Names must begin with a letter, a \$ sign or an underscore (_).
- Names are **case sensitive** (X is different from x).
- **Reserved words** (JavaScript keywords) cannot be used as names.

Variable

- Named memory location used to store data.
- JS is a **dynamically typed** language.

let x=10;

const x=10; *//cannot reassign a new value*

Multiple Variable Declaration

let x=10, a=20;

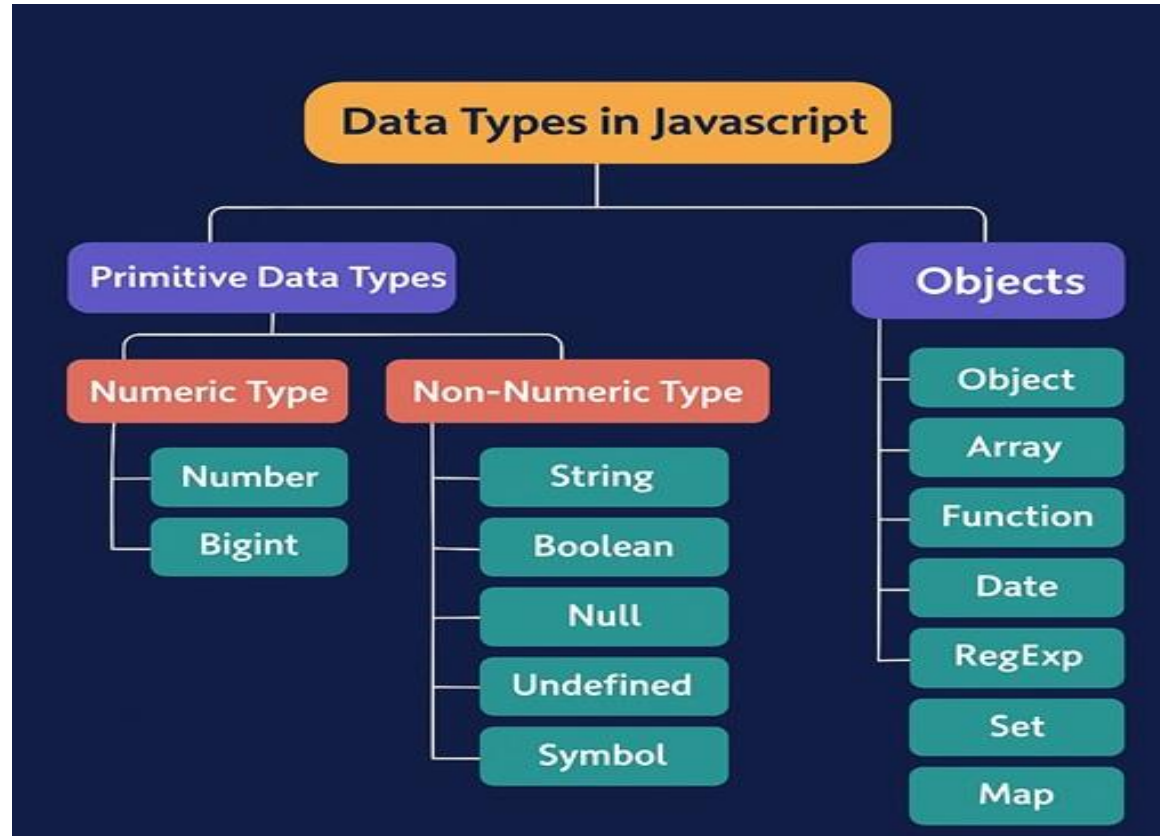
Comments: Ignored by the JS used to increase readability.

Single line comment starts with //

Multiline comments /* ... */

Data Types

- Defines what kind of value a variable can hold.
- JS has 8 main data types grouped into **Primitive & Non-Primitive Data Types**



PRIMITIVE DATA TYPES

Data Type	Description	Example
Number	Stores integers and decimal values (no separate float)	10, 3.14
String	Stores text or characters	"Hello", 'JS'
Boolean	Stores true or false	true, false
Undefined	Variable declared but not assigned	let x;
Null	Represents intentional empty value	let x = null;
BigInt	Stores very large integers, ends with n	123456789n
Symbol	Stores unique identifiers	Symbol("abc")

NON-PRIMITIVE DATA TYPES

Object	Stores collection or reference	Array, Date
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Symbol Type:

```
let x = Symbol("abc");
```

```
let b = Symbol("abc");
```

```
a === b; // false
```

- *Each call to **Symbol("abc")** creates a **unique symbol value**, even if the description is the same.*

JS Operators:

Arithmetic Operators			
Operator	Meaning	Example	Result
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division	10 / 5	2
%	Modulus	10 % 3	1
**	Exponent	2 ** 3	8
++	Increment	x++	increases x by 1
--	Decrement	x--	decreases x by 1

JS Operators:

Assignment Operators		
Operator	Meaning	Example
=	Assign	x = 10
+=	Add & assign	x += 5
-=	Subtract & assign	x -= 5
*=	Multiply & assign	x *= 5
/=	Divide & assign	x /= 5
%=	Modulus & assign	x %= 2

JS Operators:

Relational Operators

Operator	Meaning	Example	Result
==	Equal (value) Perform type coercion	5 == "5"	true
===	Strict equal (type & value)	5 === "5"	false
!=	Not equal	5 != 3	true
!==	Strict not equal	5 !== "5"	true
>	Greater than	10 > 5	true
<	Less than	5 < 10	true
>=	Greater or equal	5 >= 5	true
<=	Less or equal	5 <= 3	false

JS Operators:

Logical Operators			
Operator	Meaning	Example	Result
&&	AND	true && false	false
 	OR	true false	true
!	NOT	!true	false

Conditional or Ternary Operator		
Operator	Meaning	Example
?:	Shorthand for if-else	age >= 18 ? "Adult" : "Minor"

JS Operators:

Bitwise Operators			
Operator	Meaning	Example	Result
&	Bitwise AND	5 & 1	1
	Bitwise OR	3 2	3
^	Bitwise XOR	5 ^ 1	4
~	Bitwise NOT	~5 ~n= -(n+1)	-6
<<	Left shift	5 << 1	10
>>	Right shift	5 >> 1	2

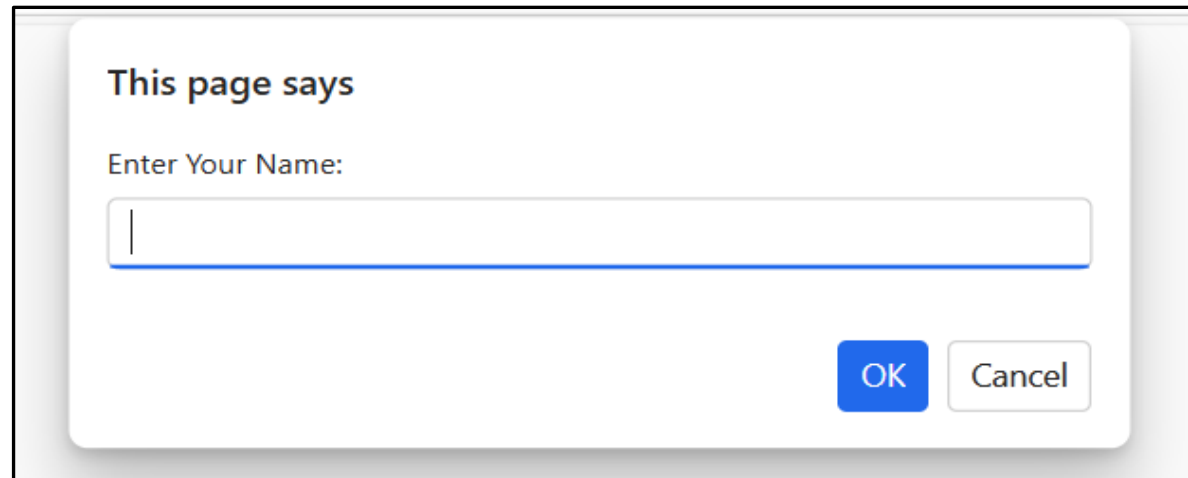
JS Operators:

Type Operators			
Operator	Meaning	Example	Result
typeof	Used for Type Checking	typeof 10	"number"
instanceof	Object type <i>Checks if an object is an instance of a specific constructor or class</i>	Let arr=[1,2] arr instanceof Array;	true

prompt():

- Used to read input from the user.
- Display a dialog box to the user & returns user input as a string.

let input = prompt("Enter Your Name:");



Note:

- *Input is always a **string***
- *Use **parseInt()** or **parseFloat()** to convert to a number*